Unofficial English translation, made for demonstration purposes. Layout simplified; this is not the original document. Original: tijhuisingenieurs.nl/files/tijhuis-waterkwaliteit-en-biodiversiteit.pdf

www.tijhuisingenieurs.nl

More information about water quality and biodiversity can be found on our website:

Water quality and biodiversity

Many waters still do not meet the standards imposed under the WFD. This is often caused by excessive nutrients in the water. The water receives these nutrients, among other sources, from the water bed, where the exchange of nutrients takes place. In nutrient-rich water, algae and duckweed have plenty of room to grow. If invasive exotic species are also present, valuable native (underwater) vegetation is displaced. Water quality and biodiversity go hand in hand.

The nutrient content of the water bed therefore influences the quality of the water and biodiversity. By taking this into account in regular maintenance, the exchange can be limited. This contributes to improving water quality and increasing biodiversity.

Integrated approach to water quality and biodiversity

With a small extension of the regular survey, a statement can be made about the water quality. The link is then made between nutrient sources that are still present (combined sewer overflows, heavily fertilised plots, bird droppings) and the nutrients in the sludge layer. The strategy for improving water quality is to remove or cover the nutrient-rich contact layer and to create nature-friendly banks or restore legakkers. When dredging, the underlying "new contact layer" must then be cleaner than the previous sludge layer. Removing the nutrient-rich layer every 10 years already makes a considerable difference. You have to start somewhere when improving quality. If dredging does not offer a solution, covering the water bed may still be an option.

Services and products

Drawing up an opportunities and bottlenecks map: the water passport

Analysis of nutrients in the sludge layer, solid bed and pore water

Measuring physical parameters such as transparency, pH and water temperature

Ecological quick scan of the water system

Design and layout of nature-friendly banks and restoration of legakkers

Optimisation of the dredging strategy to improve water quality

Approach